are to open a circuit under short circuit or overload conditions.

i. It is used in higher current ratings when compared to miniature circuit breakers.
ii. The breaking current ratings of a moulded circuit breaker can be modified.

8.7.1 Comparison between miniature circuit breaker and moulded case circuit breaker

Sl. No.	Miniature circuit breaker	Moulded case circuit breaker
1	Suitable for electrical circuit less than 100 Amperes	Suitable for electrical circuit between 63 to 3000 Amperes
2	It protects the electrical circuit from short circuit fault and over load	It protects the electrical circuit from short circuit fault and over load
3	Mostly used in homes	Used in industries, commercial buildings and hospital

8.8 Earth Leakage Circuit Breaker (ELCB)

The ELCB is used to protect the circuit from the electrical leakage. When someone gets an electric shock, then this circuit breaker cuts off the power at the time of 0.1 second for protecting the personal safety and avoiding the fear from the short circuit and overload.

ELCB is a security device used in electrical system with high earth impedance to avoid shock. The main principle of earth leakage protector is to stop injury to humans and nature due to electric shock.

In figure 8.9, the ELCB is connected between the conductor and earth. The ELCB notices fault currents from live to the ground wire inside the installation it guards. If enough voltage emerges across the search coil in the circuit breaker, it will turn off the supply, and stay off until reset by manual.

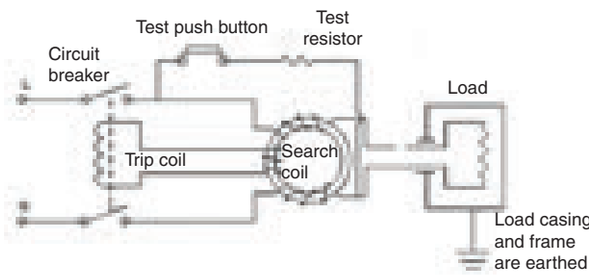


Fig. 8.9 Earth leakage circuit breaker

8.8.1 Types of earth leakage circuit breaker

1. Voltage operated earth leakage circuit breaker (Voltage – ELCB)
2. Current operated earth leakage circuit breaker (Current – ELCB)

8.8.2 Voltage operated earth leakage circuit breaker

One terminal of the relay coil is connected to the metal body of the equipment to protect against earth leakage

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